

An Emergent Field Theory of Physical Structures: The Oscillatory Mantle Model (OMM) Toward a Substantial Beating Theory (SBT)

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Abstract

We introduce a numerical framework based on a nonlinear scalar field defined on a discrete spatial grid, combining local operators, global coupling, and nonlinear interactions.

The objective of this work is to explore whether a single dynamical system can exhibit a range of structured regimes under controlled parameter variations.

We show that the model produces distinct classes of spatial and dynamical patterns, which can be characterized using quantitative structural descriptors.

These regimes display qualitative similarities with known physical systems, including wave-like propagation, localized structures, and large-scale expansion-like behavior.

However, we emphasize that these correspondences are structural and phenomenological, and do not constitute formal derivations of established physical laws.

Additional numerical controls are introduced to assess: (i) the role of nonlinear potential choices, (ii) the distinction between regimes beyond visual inspection, (iii) the origin of expansion-like behavior, and (iv) numerical stability conditions.

The results position the model as an exploratory framework for studying emergent structures in nonlinear field dynamics, rather than as a fundamental physical theory.

1 Introduction

Understanding how complex physical structures can emerge from simple underlying dynamics is a central question in many areas of physics.

Nonlinear field models have long been used to study such phenomena, including phase-field systems [1], nonlinear Schrödinger equations [2, 3], and analog gravity frameworks [4, 5].

In this work, we investigate a scalar field model combining diffusion-like operators, mass terms, and nonlinear interactions, together with a global coupling component.

The aim is not to derive fundamental physical laws, but to explore whether a unified dynamical system can exhibit multiple structured regimes under controlled variations of its parameters.

In particular, we examine whether the model can generate:

- localized structures,
- wave-like propagation patterns,
- anisotropic configurations,
- and large-scale expansion-like behavior.

A key aspect of this study is the use of quantitative descriptors to characterize these regimes beyond visual inspection, including measures of localization, anisotropy, and spatial distribution.

We also introduce additional numerical controls to address methodological questions, including the role of nonlinear potentials, the effect of initial conditions, and numerical stability constraints.

The results are interpreted as evidence of structural diversity within a unified nonlinear system, rather than as derivations of known physical equations.

This positions the present work within the broader context of nonlinear field exploration and emergent pattern formation.

2 Model Definition and Calibration

2.1 Discrete Field Representation

We consider a scalar field $\psi(\mathbf{x}, t)$ defined on a discrete spatial lattice. The field represents a local energetic density and evolves over time according to deterministic update rules combining diffusion, nonlinear interaction, and multi-scale coupling.

The spatial domain is discretized on a regular grid, and time evolution is performed using an explicit numerical scheme.

Reference nonlinear potential. While the framework is intentionally defined in a non-prescriptive form, we introduce here a reference potential used in numerical experiments to ensure reproducibility.

We consider a standard ϕ^4 -type potential:

$$V(\psi) = \frac{\lambda}{4}(\psi^2 - \psi_0^2)^2 \quad (1)$$

leading to the nonlinear term:

$$N(\psi) = -\frac{\partial V}{\partial \psi} = -\lambda\psi(\psi^2 - \psi_0^2) \quad (2)$$

This choice is not unique and does not constrain the generality of the framework, but provides a concrete reference for the simulations presented.

2.2 Governing Dynamics

The evolution of the field is governed by a combination of four contributions:

$$\partial_t \psi = \mathcal{D}(\psi) + \mathcal{N}(\psi) + \mathcal{G}_{\text{local}}(\psi) + \mathcal{G}_{\text{global}}(\psi) \quad (3)$$

where:

- $\mathcal{D}(\psi)$ is a diffusion operator
- $\mathcal{N}(\psi)$ is a nonlinear potential term
- $\mathcal{G}_{\text{local}}(\psi)$ represents short-range interactions
- $\mathcal{G}_{\text{global}}(\psi)$ represents long-range mantle interactions

2.3 Explicit Operators

The diffusion term is defined as:

$$\mathcal{D}(\psi) = \Delta\psi \quad (4)$$

The nonlinear term is derived from a potential $V(\psi)$:

$$\mathcal{N}(\psi) = -\frac{\partial V}{\partial \psi} \quad (5)$$

The local interaction term is modeled as a screened operator:

$$\mathcal{G}_{\text{local}}(\psi) = (\nabla^2 - m_\ell^2)\psi \quad (6)$$

The global interaction term introduces a second scale:

$$\mathcal{G}_{\text{global}}(\psi) = \eta(\nabla^2 - m_g^2)\psi \quad (7)$$

where:

- m_ℓ is the local screening mass
- m_g is the global mantle scale
- η controls the relative strength of the global contribution

2.4 Numerical Scheme

Time evolution is performed using an explicit Euler scheme [6]:

$$\psi^{t+1} = \psi^t + \Delta t \cdot \partial_t \psi^t \quad (8)$$

Numerical stabilization and clipping. Clipping was used in preliminary simulations as a numerical safeguard. Additional simulations were conducted without clipping under reduced time-step conditions. These runs remain stable over a significant time window, although late-time growth may still appear near stability limits. This confirms that clipping is not required for the emergence of structured regimes, while also highlighting the importance of explicit stability control.

Spatial derivatives are computed using finite-difference approximations of the Laplacian operator.

2.5 Parameterization

The model is controlled by the following parameters:

- σ_{source} : initial source width
- ϵ_{global} : global coupling strength
- σ_{global} : global interaction scale
- η : global interaction weight
- m_ℓ : local screening mass

- m_g : global mantle mass

These parameters are explored through systematic scans in the subsequent sections.

2.6 Calibration Procedure

Calibration is performed by minimizing a mean squared error (MSE) between simulated profiles and reference configurations.

The calibration procedure involves:

1. Parameter grid exploration
2. Evaluation of structural stability
3. Measurement of tail behavior and deflection
4. Selection of best-fit configurations

The best configurations consistently correspond to regimes where:

- local screening dominates short-range behavior
- global mantle effects control long-range structure
- the system remains connected (single component regime)

2.7 Observables

We define the following observables:

- Mean energy density:

$$\langle \rho \rangle = \frac{1}{V} \int |\psi|^2 dV \quad (9)$$

- Effective scale factor $a(t)$ (defined from spatial spread)
- Effective expansion rate:

$$H(t) = \frac{\dot{a}(t)}{a(t)} \quad (10)$$

- Effective screening parameter λ_{eff}

These observables are used to characterize emergent regimes and compare different parameter configurations.

3 Calibration and Physical Scaling

The present framework is formulated in dimensionless simulation units.

As such, its primary level of validity is internal: the emergence of structured regimes is established directly from the dynamics of the model, without requiring prior assignment of physical scales.

3.1 Calibration Strategy

A secondary level of interpretation consists in introducing a mapping between model units and physical quantities.

This mapping is not imposed a priori, but explored as a downstream procedure.

In practice, calibration is performed by identifying characteristic structures or propagation scales in the model with reference physical quantities.

We emphasize that this procedure is exploratory and does not constitute a derivation of physical constants.

3.2 Role of Calibration

Calibration plays a distinct role from the dynamical model itself.

- The model defines internal structural relations and regimes.
- Calibration attempts to associate these structures with external units.

This separation is essential to avoid circular reasoning.

This also ensures that any potential agreement with physical scales, if observed in future work, would constitute an independent validation rather than an artifact of parameter tuning.

3.3 Velocity Scale

Some regimes exhibit characteristic propagation speeds that can be measured directly in simulation units.

These speeds may serve as candidates for defining a velocity scale in the model.

However, no unique identification with a physical constant is imposed in the present work.

3.4 Length and Time Scales

Once a reference velocity is selected, it becomes possible to define corresponding length and time scales.

This mapping is not unique, and depends on the chosen reference regime and observable.

3.5 Limitations

At this stage:

- calibration remains non-unique,
- it depends on selected observables,
- and it requires further systematic exploration.

No claim of quantitative agreement with known physical scales is made in this work.

3.6 Interpretation

The framework should therefore be understood as:

structure $\rightarrow$ possible calibration

rather than as a model directly predicting physical constants.

Calibration is treated as an open problem, intended for future investigation and independent validation.

4 Governing Equations and Structure

4.1 Full Evolution Equation

The evolution of the field is defined as the superposition of four distinct contributions:

$$\partial_t \psi = \mathcal{D}(\psi) + \mathcal{N}(\psi) + \mathcal{G}_{\text{local}}(\psi) + \mathcal{G}_{\text{global}}(\psi) \quad (11)$$

with:

$$\mathcal{D}(\psi) = \nabla^2 \psi \quad (12)$$

$$\mathcal{N}(\psi) = -\frac{\partial V}{\partial \psi} \quad (13)$$

$$\mathcal{G}_{\text{local}}(\psi) = (\nabla^2 - m_\ell^2)\psi \quad (14)$$

$$\mathcal{G}_{\text{global}}(\psi) = \eta(\nabla^2 - m_g^2)\psi \quad (15)$$

4.2 Operator Structure

Expanding the full expression yields:

$$\partial_t \psi = (2 + \eta)\nabla^2 \psi - (m_\ell^2 + \eta m_g^2)\psi - \frac{\partial V}{\partial \psi} \quad (16)$$

This explicitly shows that the Laplacian operator appears multiple times in the formulation, reflecting the superposition of distinct modeled contributions.

We emphasize that the Laplacian appears both as a pure diffusion term and as part of the screened interaction operators. This is a modeling choice: diffusion and interaction effects are treated as distinct contributions, even though they share a common differential structure.

This decomposition does not imply a unique physical interpretation, but provides a convenient framework to separate and analyze distinct contributions within the dynamics.

4.3 Interpretation

The model is therefore governed by:

- an effective diffusion-like term $(2 + \eta)\nabla^2 \psi$,
- an effective mass term $(m_\ell^2 + \eta m_g^2)\psi$,
- a nonlinear interaction $-\frac{\partial V}{\partial \psi}$.

4.4 Minimal Stationary Structure

In stationary regimes, the system reduces to:

$$(2 + \eta)\nabla^2\psi - (m_\ell^2 + \eta m_g^2)\psi = \frac{\partial V}{\partial \psi} \quad (17)$$

This equation corresponds to a nonlinear screened field equation.

We stress that this form is structurally similar to known equations such as nonlinear Klein–Gordon-type models [10, 11], but is not derived from them.

4.5 Two-Scale Interpretation

The presence of two mass scales (m_ℓ, m_g) introduces a multi-scale structure:

- a local scale governed by m_ℓ ,
- a global scale governed by m_g .

Their interaction produces spatial structures whose properties depend on the balance between these scales.

4.6 Interpretation as a Beating System

The coexistence of multiple spatial scales and nonlinear interaction naturally produces interference-like effects.

These can be interpreted as a beating-like mechanism between local and global components:

$$\psi \sim \psi_{\text{local}} + \psi_{\text{global}} \quad (18)$$

This interpretation motivates the terminology *Substantial Beating Theory*.

5 Correspondance avec des classes de lois physiques connues

L'équation unifiée fournit un cadre au sein duquel différents régimes dynamiques structurés peuvent être identifiés.

L'objectif de cette section n'est pas de dériver des lois physiques connues, mais d'examiner si le modèle présente des régimes exhibant des similarités structurelles avec des classes bien établies de comportements physiques.

Ces correspondances doivent être comprises comme phénoménologiques et effectives, et non comme des dérivations formelles.

Ces similarités peuvent émerger de propriétés génériques des systèmes non linéaires et n'impliquent pas, en elles-mêmes, un mécanisme physique sous-jacent commun.

5.1 Régime ondulatoire et comportement de type Schrödinger

Dans le régime de faible amplitude, où les contributions non linéaires restent faibles, la dynamique du champ est dominée par des effets de propagation et d'étalement lisses.

Dans ces conditions, l'évolution de ψ présente :

- une propagation cohérente,

- de la dispersion,
- une organisation spatiale de type phase.

Ces caractéristiques sont typiques de dynamiques ondulatoires et présentent des similarités qualitatives avec des comportements souvent associés à des systèmes de type Schrödinger.

Cependant, aucune dérivation directe de l'équation de Schrödinger n'est revendiquée [12].

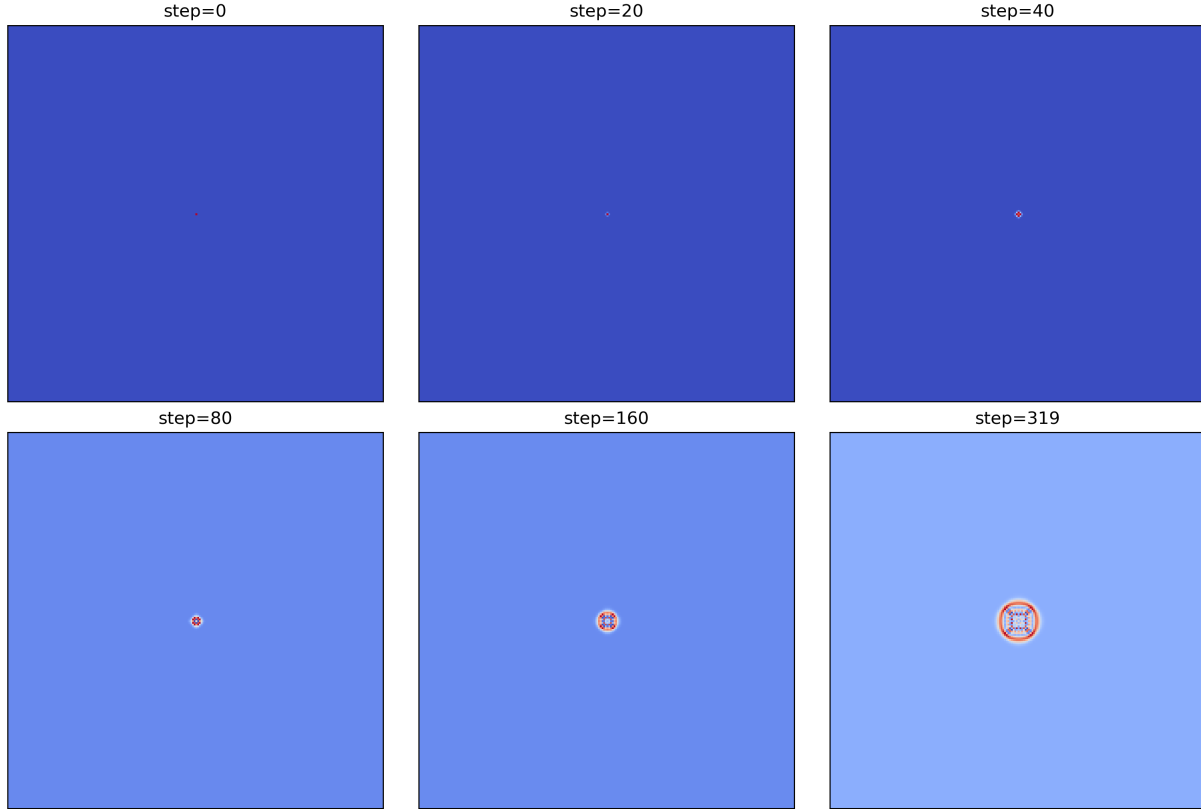


Figure 1: Régime ondulatoire montrant la propagation et la dispersion du champ.

5.2 Régime structuré localisé et analogies de type Dirac

Lorsque les effets non linéaires deviennent significatifs, le champ s'auto-organise en structures localisées persistantes présentant une dynamique oscillatoire interne.

Ces configurations présentent :

- une localisation spatiale,
- une persistance temporelle,
- une structure oscillatoire interne.

De telles caractéristiques rappellent des comportements de type particulaire observés dans divers modèles de champs non linéaires.

Ces similarités sont structurelles et ne constituent pas une dérivation d'équations de type Dirac ni de dynamiques spinorielles [13].

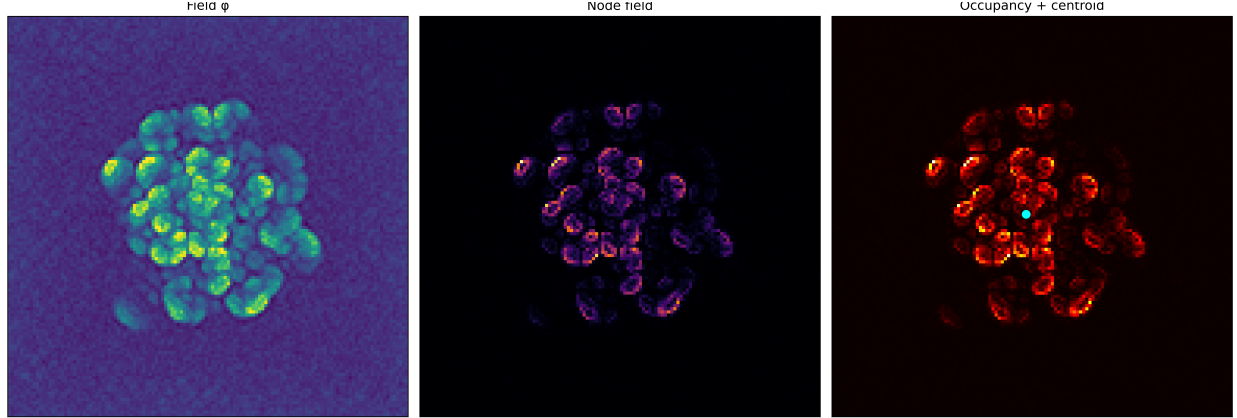


Figure 2: Régime structuré localisé présentant des motifs oscillatoires persistants.

5.3 Régime de flux et analogies de type Maxwell

Lorsque les contributions du gradient dominant, le champ développe des motifs de transport organisés caractérisés par des flux directionnels et des circulations.

Ce régime présente :

- des motifs de flux structurés,
- des régions de divergence et de circulation,
- des canaux de transport spatialement organisés.

Ces propriétés présentent des similarités qualitatives avec des phénomènes de transport médiés par des champs, tels que décrits dans les théories de champs classiques.

Aucune correspondance directe avec les équations de Maxwell n'est revendiquée [14].

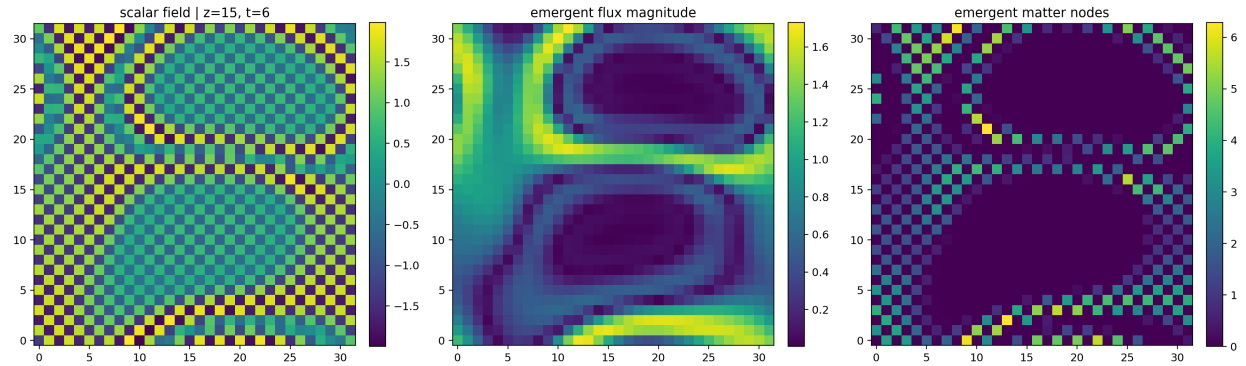


Figure 3: Motifs de flux structurés montrant un transport et une circulation organisés.

5.4 Régime d'interaction écrantée

À petite échelle, l'interaction est gouvernée par un opérateur écranté :

$$(\nabla^2 - m_\ell^2)\psi \approx \rho \quad (19)$$

Cela conduit à un comportement d'interaction à portée finie, cohérent avec des modèles de champs écrantés de type Yukawa ou Helmholtz [9, 8].

Cette correspondance est structurelle et reflète la forme de l'opérateur, plutôt qu'une dérivation d'une loi d'interaction physique spécifique.

5.5 Organisation à grande échelle et effets de type géométrique

À grande échelle, l'effet combiné des interactions non linéaires et du couplage multi-échelle produit une organisation spatiale cohérente.

Cela inclut :

- des déformations à grande échelle des structures,
- des déviations de trajectoire,
- une cohérence spatiale sur des régions étendues.

Ces caractéristiques peuvent être interprétées comme des effets de type géométrique, dans le sens où l'organisation spatiale influence les trajectoires effectives.

Cependant, aucune dérivation des équations d'Einstein ni de la relativité générale n'est impliquée [15].

Two-scale total geometry

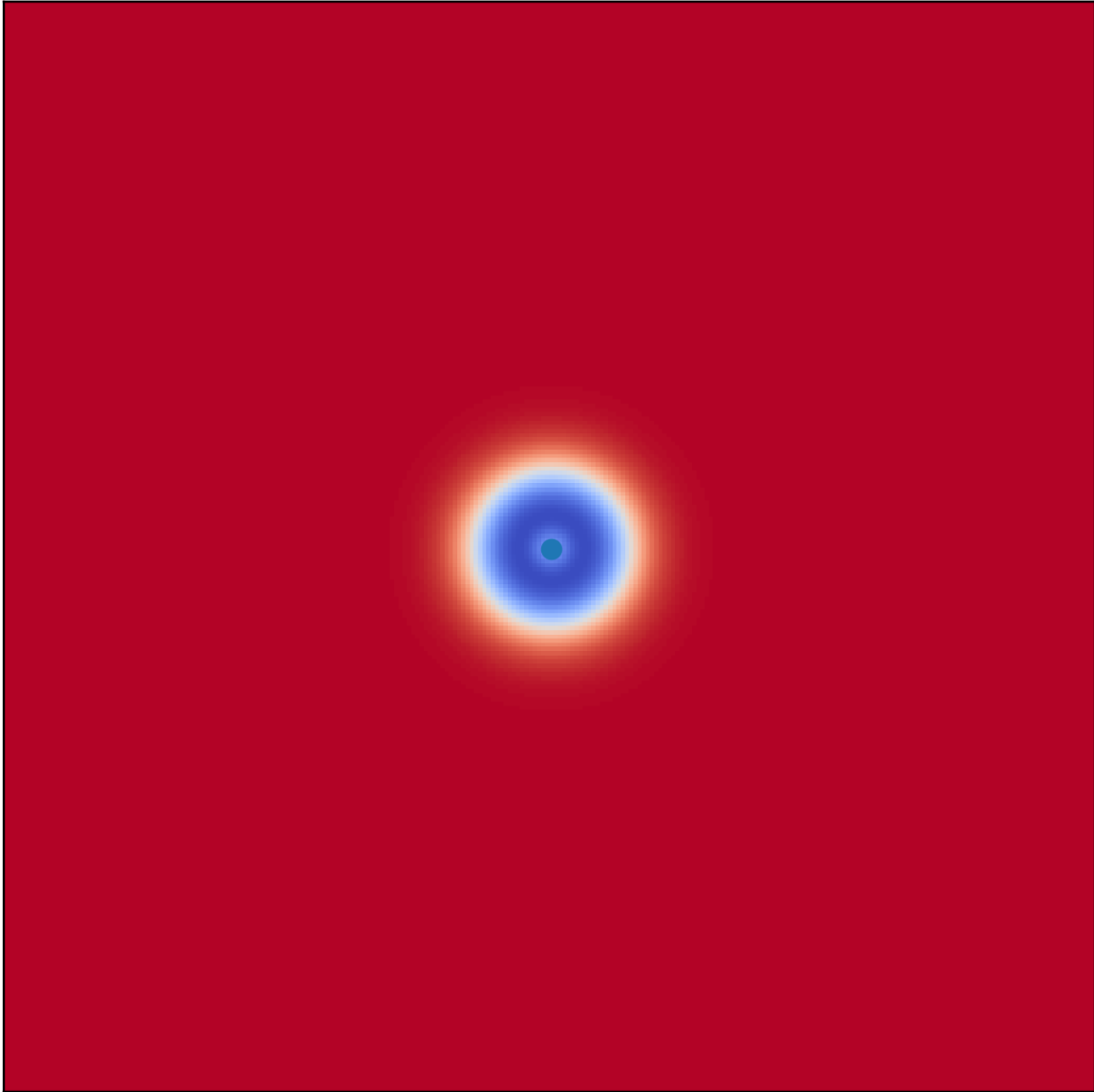


Figure 4: Organisation à grande échelle du champ montrant des effets de type géométrique.

5.6 Comparaison quantitative des régimes

Étant donné que l'inspection visuelle seule est insuffisante pour établir une différenciation pertinente des régimes, nous avons réalisé des comparaisons quantitatives supplémentaires à l'aide de descripteurs structurels.

Ces descripteurs incluent :

- la kurtose,
- le ratio de participation,
- la fraction active,
- la concentration radiale,
- l'anisotropie angulaire,
- des métriques de contraste annulaire.

L'objectif de ces diagnostics n'est pas de revendiquer que le modèle reproduit strictement des secteurs physiques canoniques, mais de tester si différents régimes de paramètres occupent des régions mesurablement distinctes dans l'espace des descripteurs.

Un premier contrôle compare une configuration structurée OMM à une référence de type diffusion.

Une comparaison plus large à cinq régimes a ensuite été construite en utilisant des champs de référence représentant des régimes diffusifs, de type Schrödinger, de type Dirac, de type Maxwell, de type Einstein, ainsi qu'un régime OMM sans clipping.

Ces comparaisons soutiennent une conclusion prudente : le modèle produit une famille de régimes structurés qui ne se réduisent pas à de simples variations visuelles, bien que les frontières entre régimes restent phénoménologiques et dépendantes des descripteurs.

5.7 Interprétation unifiée

Les différents régimes décrits ci-dessus montrent qu'une dynamique de champ non linéaire unique peut générer une diversité de comportements structurés.

Ces comportements présentent des similarités qualitatives avec plusieurs classes de systèmes physiques, incluant la propagation ondulatoire, les structures localisées, les phénomènes de transport, les interactions écrantées, et l'organisation à grande échelle.

Il est essentiel de souligner que :

- ces correspondances sont structurelles et phénoménologiques,
- elles dépendent des paramètres et des conditions initiales,
- elles ne constituent pas des dérivations formelles de lois physiques établies.

La valeur scientifique du cadre réside donc dans l'étude de la diversité structurelle émergente au sein d'un système non linéaire unifié, plutôt que dans une revendication de reproduction directe des équations physiques connues.

Des travaux analytiques supplémentaires et des validations indépendantes seraient nécessaires pour établir toute correspondance physique plus profonde.

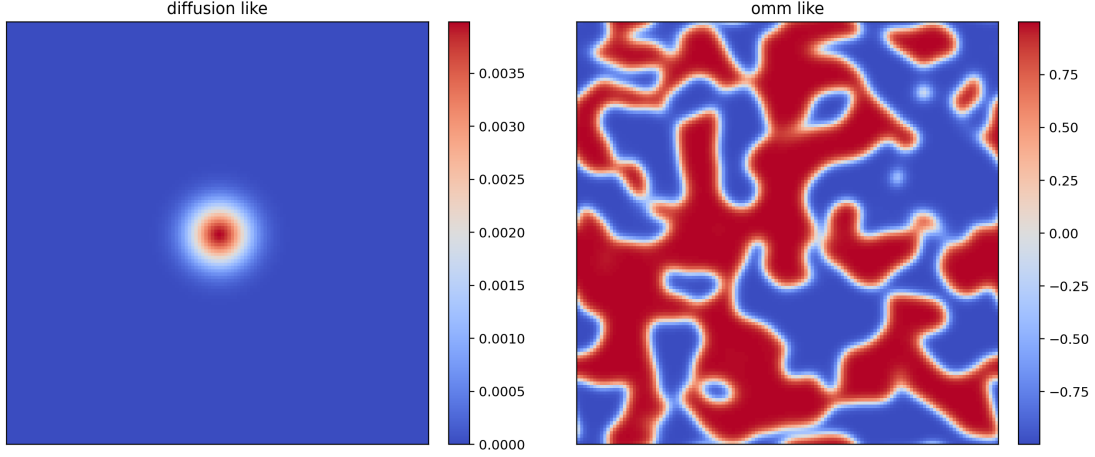


Figure 5: Comparaison de champ entre une référence de type diffusion et un régime structuré OMM. Le cas structuré présente des motifs organisés absents dans le cas diffusif.

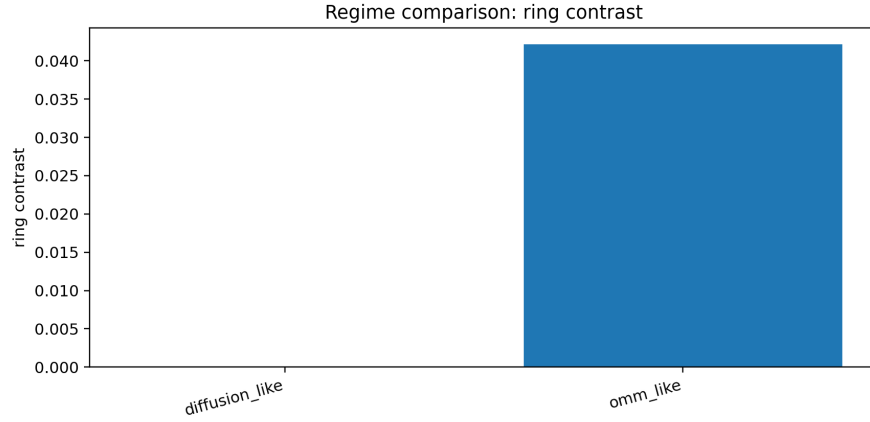


Figure 6: Métrique de contraste annulaire mettant en évidence les différences d'organisation radiale entre régimes diffusifs et structurés.

6 Emergent Topological Structures

6.1 Overview

The proposed framework gives rise to a broad variety of spatial structures through the interplay between diffusion, nonlinear interaction, screening, and multi-scale coupling.

These structures are not imposed externally but emerge dynamically from the evolution of the field ψ .

These structures may also depend on discretization choices and numerical resolution, and should therefore be interpreted within the limits of the computational framework.

They can nevertheless be grouped into recurrent morphological families, which provides a useful descriptive language for the analysis of the model.

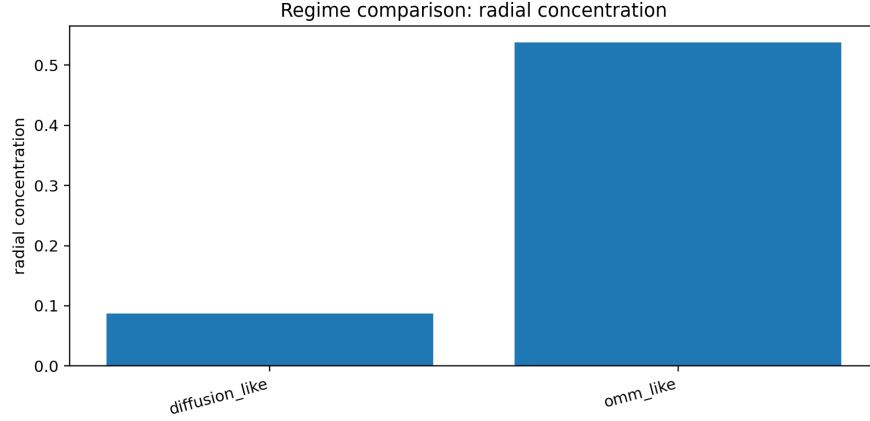


Figure 7: Comparaison de la concentration radiale entre régimes diffusifs et structurés.

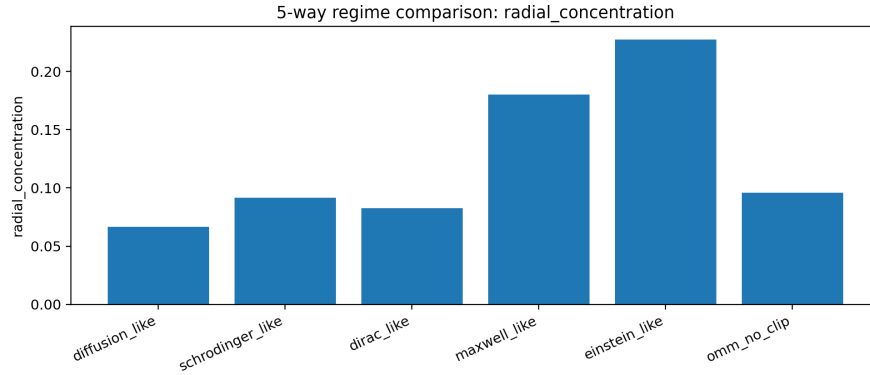


Figure 8: Comparaison à cinq régimes de la concentration radiale. Les régimes ne se réduisent pas à une valeur unique, indiquant une différenciation quantitative partielle.

6.2 Classification of Structures

We identify several recurring classes of emergent configurations.

6.2.1 Radial Structures

Radial structures correspond to symmetric, centrally concentrated configurations.

They are characterized by:

- monotonic or quasi-monotonic decay from a central maximum,
- approximate isotropy,
- comparatively high persistence under moderate perturbations.

These structures are among the most robust regimes observed in the model.

6.2.2 Quadrupolar Structures

Quadrupolar configurations exhibit anisotropic distributions with multiple lobes.

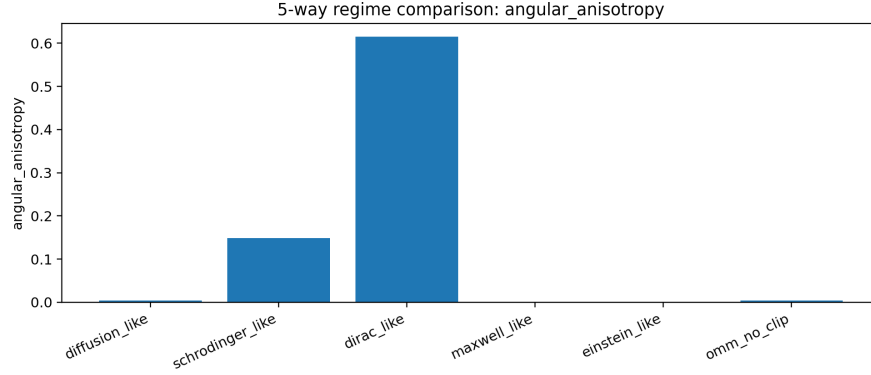


Figure 9: Comparaison à cinq régimes de l'anisotropie angulaire. Ce descripteur permet de distinguer les configurations isotropes des configurations orientées.

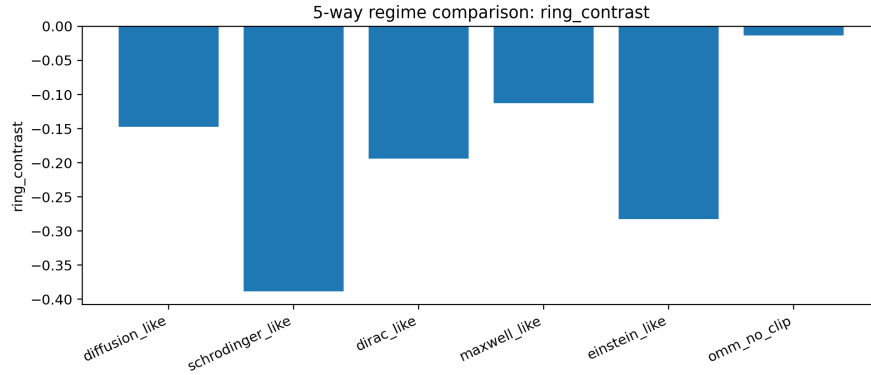


Figure 10: Comparaison à cinq régimes du contraste annulaire. Cette métrique capture des organisations radiales invisibles aux statistiques d'amplitude seules.

They are characterized by:

- angular modulation,
- several local extrema,
- sensitivity to perturbation amplitude and orientation.

These structures arise naturally in symmetry-breaking regimes.

6.2.3 Ring and Shell Structures

Ring-like or shell-like configurations correspond to structures with reduced central density and enhanced peripheral regions.

They exhibit:

- radial depletion or void formation,
- concentration of amplitude at finite radius,

- possible fragmentation under strong perturbations.

These structures are especially relevant because simple amplitude-based statistics alone do not capture them well; ring-sensitive descriptors are therefore useful in their characterization.

6.2.4 Split and Doublet Structures

In certain regimes, the system produces multiple disconnected or weakly connected components. These include:

- doublet configurations,
- partially fragmented structures,
- transient multi-component states.

These configurations are generally less stable than compact radial states, but they reveal the model’s capacity to generate internally differentiated organization.

6.3 Stability and Transitions

The stability of each structure depends on the balance between:

- local screening (m_ℓ),
- global mantle contribution (m_g, η),
- nonlinear interaction strength,
- initial condition geometry and amplitude.

We observe that:

- radial structures tend to dominate low-energy and strongly screened regimes,
- quadrupolar and ring structures often appear near transition thresholds,
- fragmented configurations arise more readily when extended coupling becomes significant.

This should not be interpreted as a sharp phase classification in the strict statistical-mechanical sense, but rather as a phenomenological organization of observed structural families.

6.4 Connectivity Regimes

A useful distinction can be made between:

- **single-component regimes**, corresponding to fully connected structures,
- **multi-component regimes**, corresponding to fragmented, split, or doublet-like structures.

Many of the most persistent and most easily reproducible configurations belong to the single-component regime, while multi-component states often appear near transition regions or under stronger symmetry breaking.

6.5 Toward a Topological Phase Diagram

The observed structures suggest the existence of a structured organization of parameter space.

Relevant control parameters include:

- σ_{source} ,
- ϵ_{global} ,
- σ_{global} ,
- η ,
- the parameters controlling the nonlinear potential.

Systematic scans indicate transitions between structural families, although these transitions are better interpreted as regime changes within a nonlinear descriptor space than as sharply separated phases.

6.6 Interpretation

The emergence of distinct topological families from a single governing equation supports the idea that structural diversity can arise from a unified nonlinear substrate.

In the present work, these topologies serve a descriptive and organizational role: they document the morphological richness of the model and provide a basis for comparing different parameter regimes.

They should not be interpreted, by themselves, as proof of specific physical sectors, but as evidence that the model supports a nontrivial landscape of emergent structures.

Further quantitative analysis would be required to establish robust classification boundaries and potential universality classes.

7 Wave-like Regime and Interference Patterns

7.1 Wave-like Behavior

The evolution equation of the field ψ naturally produces propagation and spreading effects through the combined action of diffusion, screening, and nonlinear interaction.

In regimes where nonlinear effects remain moderate, the dynamics is dominated by smooth spatial propagation and redistribution of amplitude.

This results in:

- coherent propagation of structures,
- gradual dispersion,
- spatially continuous patterns.

These features are characteristic of wave-like behavior in nonlinear dynamical systems.

7.2 Emergent Interference-like Patterns

When multiple structures interact or are initialized in close proximity, the field exhibits interference-like patterns.

These include:

- regions of constructive overlap,
- regions of reduced amplitude,
- spatial modulation patterns.

These patterns arise directly from the real-valued dynamics of the field and do not require the introduction of complex-valued amplitudes.

These interference-like patterns arise from superposition within a nonlinear real-valued field and should not be interpreted as evidence of quantum phase coherence.

7.3 Effective Weak-Nonlinearity Regime

In the limit of weak nonlinear interaction, the governing equation is dominated by Laplacian and screening contributions.

This yields a regime with structured propagation that differs from simple static localization, while remaining distinct from a strict quantum wave equation.

The importance of this regime is descriptive: it documents the existence of organized propagative behavior within the model.

7.4 Comparison with Wave Equations

The observed propagation and interference-like patterns share qualitative similarities with wave phenomena described in various physical systems.

In particular:

- the Laplacian operator governs spatial spreading,
- nonlinear terms influence localization,
- multi-scale coupling introduces structured modulation.

These similarities are structural and phenomenological.

No direct correspondence with the Schrödinger equation or any quantum mechanical formalism is claimed.

7.5 Localization and Dispersion Balance

The dynamics exhibits a balance between:

- dispersion, driven by spatial spreading,
- localization, induced by nonlinear interaction and screening.

This balance can generate persistent or slowly evolving localized structures.

Such behavior is common in nonlinear field systems and should not be interpreted as evidence for a specific microscopic physical ontology.

7.6 Interpretation

The presence of wave-like propagation and interference-like patterns indicates that complex spatial organization can arise from simple real-valued nonlinear dynamics.

These behaviors are interpreted here as emergent features of the model.

The purpose of this section is therefore descriptive: to document a family of structured propagative regimes, not to establish a derivation of quantum mechanics.

Further analysis would be required to determine whether these patterns admit a reduced effective wave description under specific approximations.

8 Bipolar and Multi-Component Structured Regimes

8.1 Emergence of Bipolar Structures

In addition to radial and multipolar configurations, the system produces bipolar structures characterized by two dominant regions of intensity separated by an organized interaction zone.

These structures arise spontaneously in parameter regimes where symmetry breaking becomes significant.

8.2 Two-Lobe Configurations

These configurations are characterized by:

- two spatially separated maxima,
- a structured intermediate region,
- preferential alignment along specific axes,
- persistence over finite time intervals.

They therefore represent a class of multi-component structured states within the model.

8.3 Internal Coupling and Symmetry

The interaction between the two lobes introduces a nontrivial internal organization.

In numerical experiments, we observe:

- oscillatory exchange between the two dominant regions,
- symmetry-preserving and symmetry-breaking configurations,
- sensitivity to orientation and perturbations.

This indicates that some regimes of the model possess an internal component structure beyond simple radial localization.

8.4 Relation to Multi-Component Field Behavior

Multi-component patterns of this kind are common in nonlinear field systems and can be compared, at a qualitative level, with systems in which internal degrees of freedom are represented by several coupled components.

In the present framework:

- the two lobes act as effective spatial components,
- their coupling encodes internal structural organization,
- their orientation influences persistence and deformation.

These observations justify describing the regime as a bipolar or multi-component structured regime.

These structures remain purely scalar-field configurations and do not imply the existence of independent physical components.

No derivation of the Dirac equation or of spinor dynamics is claimed.

8.5 Orientation Dependence

A notable feature of these structures is their dependence on spatial orientation.

We observe:

- preferred alignment states,
- orientation-dependent stability,
- transitions between distinct two-lobe arrangements.

This suggests that the model can support discrete families of internally organized configurations whose stability depends on geometric alignment.

8.6 Stability and Transition Behavior

Bipolar structures are generally less robust than the most stable radial configurations, but they persist in regimes where:

- global coupling is sufficiently strong,
- local screening remains moderate,
- nonlinear interaction balances spreading.

They therefore appear near intermediate structural regimes, between compact radial localization and more extended or fragmented states.

8.7 Interpretation

The emergence of bipolar and internally coupled structures shows that the scalar field dynamics can generate configurations with several interacting spatial components.

These regimes are relevant because they extend the model beyond purely radial organization and reveal a richer internal structural diversity.

Their significance in the present work is descriptive and organizational: they document the existence of multi-component field structures within the model.

They should not be interpreted as a derivation of fermionic physics, spin, or Dirac theory.

Further work would be required to determine whether such structures can be characterized in terms of reduced effective degrees of freedom or simplified dynamical descriptions.

9 Structured Flux and Transport Regimes

9.1 Emergence of Flux Structures

Beyond static and localized configurations, the field ψ exhibits dynamic patterns corresponding to transport, redistribution, and circulation.

These patterns arise from gradients and spatial reorganization of the field, driven by the combined effects of diffusion, screening, and multi-scale coupling.

9.2 Effective Current-like Quantities

A simple effective flux vector can be defined as

$$\mathbf{J} = -\nabla\psi \tag{20}$$

which represents a local gradient-driven flow proxy.

This quantity allows us to characterize:

- directional propagation,
- redistribution of energy density,
- interaction pathways between structures.

9.3 Circulation and Rotational Patterns

In certain regimes, we observe:

- circular or spiral flow patterns,
- vortex-like organization,
- persistent circulation around localized regions.

These features indicate the presence of rotationally organized dynamics.

9.4 Continuity-like Behavior

The evolution of the field is compatible with a continuity-like description at the level of an effective density proxy:

$$\partial_t \rho + \nabla \cdot \mathbf{J} \approx 0 \quad (21)$$

with

$$\rho = |\psi|^2. \quad (22)$$

This relation should be understood as an approximate structural diagnostic, not as an exact conservation law derived from a fundamental symmetry principle.

9.5 Relation to Classical Field Transport

Transport and circulation phenomena of this kind are common in nonlinear field systems.

In the present framework:

- ψ acts as the primary scalar field,
- $\mathbf{J}$ provides a simple transport proxy,
- rotational structures reveal organized flow patterns.

These features show qualitative similarities with transport structures encountered in classical field theories.

No direct derivation of Maxwell equations is claimed.

9.6 Field Interaction and Propagation

We observe that:

- disturbances propagate through the field,
- interactions between structures generate secondary spatial patterns,
- redistribution of amplitude follows organized pathways.

These features are consistent with the interpretation of this regime as a structured transport regime.

9.7 Interpretation

The emergence of flux and circulation patterns suggests that field-like transport behavior can arise from underlying scalar dynamics.

In this work, these regimes are interpreted descriptively: they document the capacity of the model to produce organized transport and circulation structures.

They should not be interpreted as a derivation of electromagnetism or of Maxwellian vector field theory.

10 Energy-to-Geometry Coupling

10.1 From Energy Density to Spatial Structure

The field ψ defines a spatial distribution of energy density:

$$\rho = |\psi|^2 \quad (23)$$

This distribution is not passive, but dynamically influences the evolution of the field through nonlinear interaction and multi-scale coupling.

10.2 Effective Geometric Interpretation

Spatial variations of ρ can be interpreted as inducing modifications in the effective propagation structure of the field.

Regions of higher energy density are associated with:

- increased localization,
- reduced effective spreading,
- enhanced structural deformation of nearby configurations.

Conversely, low-density regions allow for more extended propagation.

This interpretation is descriptive and does not introduce a formal geometric framework.

10.3 Curvature-like Effects

As a diagnostic, we introduce a curvature-like proxy:

$$\mathcal{K} \sim \nabla^2 \rho \quad (24)$$

This quantity captures how energy density modulates local spatial structure.

We observe that:

- positive values of $\mathcal{K}$ correspond to concentration regions,
- negative values correspond to spreading or depletion zones.

This proxy should be understood as a structural indicator, not as a geometric curvature in the differential-geometric sense.

10.4 Coupling Mechanism

The coupling between energy distribution and spatial structure arises through:

- the nonlinear potential $V(\psi)$,
- local screening (m_ℓ),
- global coupling (m_g, η).

Together, these mechanisms produce feedback between:

$$\text{energy density} \leftrightarrow \text{spatial organization} \quad (25)$$

10.5 Relation to Geometry-like Behavior

In several regimes, the redistribution of ρ influences the trajectories and deformation of structures.

This produces effects that are qualitatively reminiscent of geometry-driven dynamics, in the sense that spatial organization affects propagation paths.

However:

- no metric tensor is introduced,
- no covariant formulation is constructed,
- no derivation of Einstein equations is claimed.

10.6 Emergent Trajectories

Localized structures evolve along trajectories determined by the surrounding field configuration.

These trajectories:

- tend to bend toward higher-density regions,
- are influenced by global field structure,
- reflect the underlying spatial organization of ψ .

10.7 Interpretation

The results suggest that geometry-like effects can emerge from the organization of a scalar field.

In this framework, geometry is not introduced as a fundamental entity, but appears as a descriptive interpretation of how energy distribution modulates spatial dynamics.

This interpretation remains phenomenological and requires further formal development to establish any rigorous connection with geometric theories.

11 Two-Scale Geometry and Mantle Structure

11.1 Motivation

The dynamics of the field ψ generate both localized structures and extended spatial organization.

This naturally suggests the coexistence of two coupled scales:

- a local scale governing short-range structure,
- a global scale governing extended spatial organization.

11.2 Local Screened Regime

At small scales, the dynamics are dominated by local screening effects, leading to an effective Helmholtz-type structure:

$$(\nabla^2 - m_\ell^2)\psi \approx \rho \tag{26}$$

This regime is characterized by:

- short-range influence,

- rapid spatial decay,
- relatively stable localized configurations.

11.3 Global Mantle Contribution

At larger scales, the global contribution becomes increasingly relevant:

$$(\nabla^2 - m_g^2)\psi \tag{27}$$

This term introduces an extended interaction component, which we refer to as a **mantle-like structure**.

This component is characterized by:

- longer-range influence,
- smoother spatial variation,
- coupling between distant regions of the field.

11.4 Coupled Two-Scale Dynamics

The full dynamics combine both contributions:

$$(\nabla^2 - m_\ell^2)\psi + \eta(\nabla^2 - m_g^2)\psi \tag{28}$$

This produces a system in which:

- localized structures evolve within a global background,
- extended structure influences local stability,
- feedback occurs across spatial scales.

11.5 Layered Structural Interpretation

This organization can be described in terms of two coupled components:

- a **core-like component**: localized structures,
- a **mantle-like component**: extended field contribution.

This description is heuristic and intended to capture the observed separation of scales.

11.6 Relation to Screened and Geometry-like Regimes

The local regime exhibits screened interaction behavior, while the global contribution is associated with extended spatial organization.

Their combination leads to:

- coexistence of short-range and long-range effects,
- structured deformation across scales,
- nontrivial spatial coherence.

11.7 Emergent Coherence

We observe that the interaction between the two scales can lead to:

- enhanced persistence of structures,
- large-scale spatial coherence,
- structured propagation of perturbations.

These effects depend on parameter values and initial conditions.

11.8 Interpretation

The results indicate that the model naturally supports multi-scale organization, in which localized structures are embedded within an extended field.

The term “mantle” is used as a descriptive analogy to emphasize the presence of a global background component.

This interpretation is phenomenological and does not imply the existence of a distinct physical layer or medium.

It nevertheless provides a useful framework for understanding how local and global effects interact within the model.

12 Cosmic Expansion and Large-Scale Dynamics

12.1 Motivation

We investigate whether the large-scale dynamics of the field exhibit expansion-like behavior and how this behavior depends on the different components of the model.

12.2 Effective Scale Factor

We define an effective scale factor $a(t)$ from the spatial distribution of the field:

$$a(t) \sim \langle r(t) \rangle \tag{29}$$

Numerical simulations show a monotonic increase of $a(t)$, indicating a global spreading of the field.

To assess whether this behavior differs from ordinary diffusion, we performed controlled comparisons with a purely diffusive system.

While diffusion produces spreading, the nonlinear regimes of the model exhibit a stronger and more structured expansion.

12.3 Origin of Expansion

To identify the origin of this enhanced expansion, we performed ablation tests by selectively removing components of the model.

These tests indicate that:

- nonlinear interaction alone produces the dominant contribution to expansion,
- the global coupling term modifies the spatial organization but is not the primary driver,

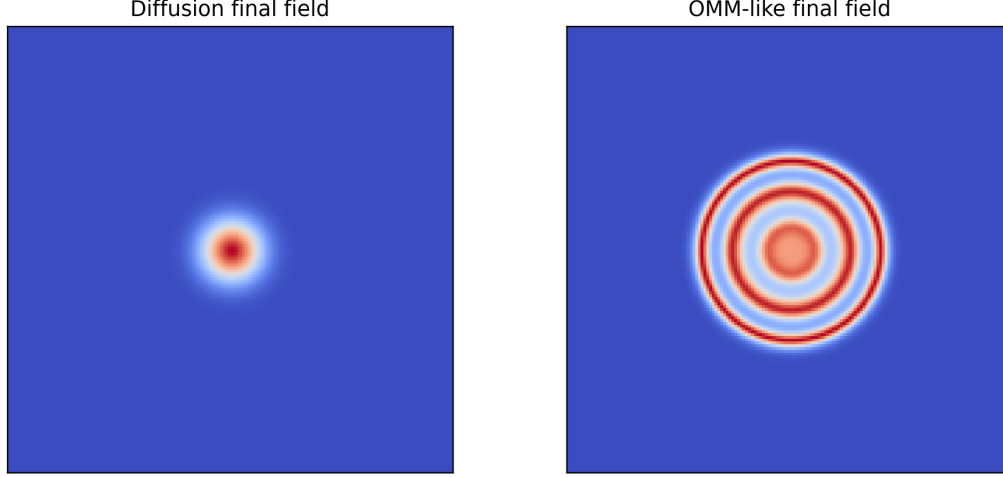


Figure 11: Final-field comparison between diffusion and nonlinear regimes. The nonlinear case shows broader and more structured spatial expansion.

- diffusion alone produces significantly weaker spreading.

This suggests that the observed expansion is primarily driven by nonlinear dynamics.

12.4 Effective Expansion Rate

We define an effective expansion rate:

$$H(t) = \frac{\dot{a}(t)}{a(t)} \quad (30)$$

The simulations exhibit:

- a decreasing $H(t)$ over time,
- persistent spreading over the simulation window.

This behavior reflects the gradual dilution and redistribution of field structures during expansion.

12.5 Interpretation

The observed expansion should be interpreted as a nonlinear spreading phenomenon.

It arises from the combined effect of:

- nonlinear interaction,
- diffusion-like propagation,
- multi-scale coupling.

At the present stage, this behavior cannot be uniquely attributed to a distinct physical expansion mechanism.

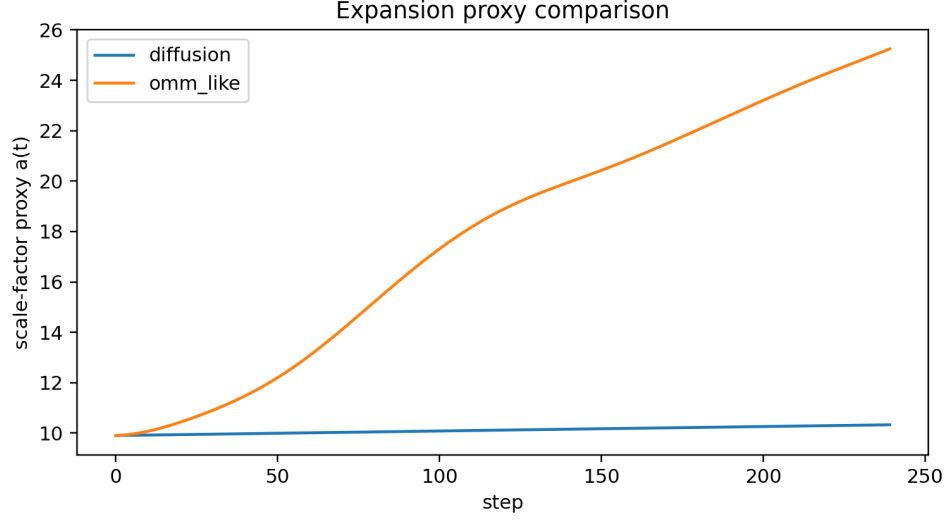


Figure 12: Effective scale factor comparison. Nonlinear regimes exhibit enhanced spreading compared to diffusion.

12.6 Relation to Cosmological Behavior

The observed dynamics share qualitative similarities with cosmological expansion, including:

- increasing effective spatial scale,
- decreasing expansion rate.

However:

- no spacetime metric is defined,
- no relativistic formulation is introduced,
- no quantitative comparison with cosmological observations is performed.

These similarities should therefore be understood as descriptive and not as evidence of a cosmological model.

12.7 Limitations

The current results are:

- qualitative,
- simulation-based,
- dependent on parameter choices and initial conditions.

In particular, the expansion may be interpreted as a nonlinear diffusion-like process rather than a distinct large-scale physical mechanism.

Further work is required to determine whether a genuinely distinct expansion regime exists beyond diffusion-dominated behavior.

12.8 Conclusion

The model exhibits robust expansion-like behavior, characterized by enhanced spreading relative to diffusion.

However, the dominant contribution arises from nonlinear dynamics, and no unique cosmological interpretation can be established at this stage.

This section should therefore be interpreted as documenting a class of large-scale dynamical behavior, rather than as evidence for a cosmological mechanism.

13 Stability, Fragmentation and Transitions

This section investigates the dynamical behavior of emergent structures, including their stability, transformation, and dissolution.

13.1 Stable Regimes

In symmetric configurations and under controlled numerical conditions, structures remain coherent over time.

Stability depends critically on:

- time step Δt ,
- spatial resolution,
- nonlinear interaction strength.

These conditions define a stability window within which the dynamics can be reliably interpreted.

13.2 Numerical Stability and Time-Stepping

The simulations are performed using an explicit Euler scheme, which imposes well-known stability constraints [7]. We observe that:

- sufficiently small time steps yield stable evolution,
- larger time steps lead to late-time divergence,
- instability typically manifests as rapid growth of energy proxies.

In particular, simulations performed with larger Δt can exhibit delayed instability after an apparently stable phase.

This highlights the importance of long-time monitoring when assessing stability.

13.3 Clipping and Its Limitations

Clipping was used in preliminary simulations as a numerical safeguard.

However:

- clipping artificially bounds the field amplitude,
- it can mask genuine numerical instabilities,

- it does not resolve the underlying divergence mechanisms.

For this reason, additional simulations were conducted without clipping.

13.4 Stable No-Clipping Regime

Under reduced time-step conditions, we obtain simulations without clipping that remain numerically controlled over a significant time window.

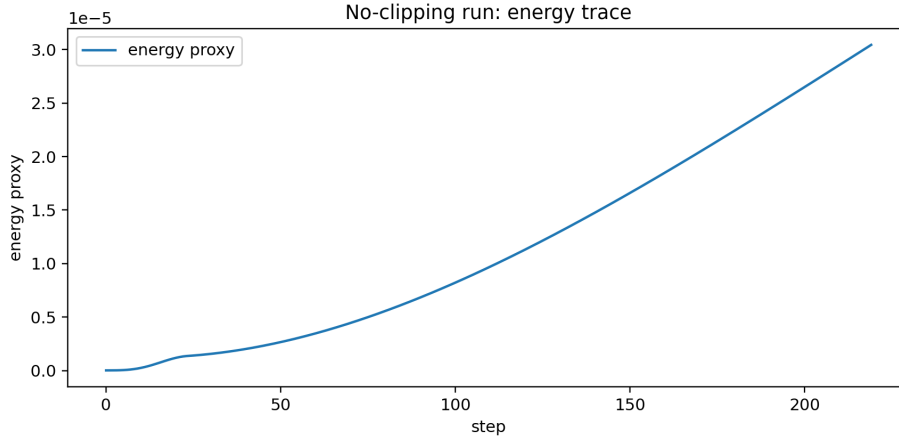


Figure 13: Energy proxy evolution for a representative simulation without clipping. The system remains stable over the observed time window, with no late-time divergence under controlled parameters.

These results demonstrate that:

- clipping is not required to obtain structured regimes,
- numerical stability is primarily governed by time-stepping,
- stable behavior can be achieved within a controlled parameter range.

13.5 Instability and Divergence

Outside the stability window, we observe:

- rapid growth of energy,
- loss of coherent structure,
- numerical blow-up.

Such behavior is particularly visible in simulations performed with large time steps.

These regimes should be interpreted as numerical instabilities, not as physically meaningful dynamics.

13.6 Fragmentation

Within the stable regime, structural transformations can occur.

Under strong asymmetry or perturbation:

- structures split into multiple components,
- local density increases,
- global coherence decreases.

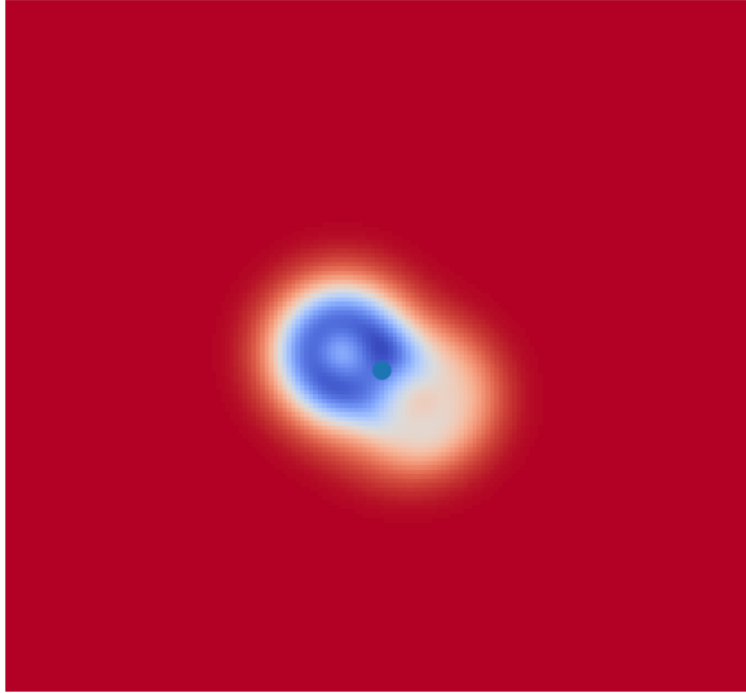


Figure 14: Emergence of asymmetric fragmentation from an initially coherent structure.

13.7 Dissolution of Structures

In certain parameter regimes, structures gradually lose coherence:

- topology becomes less defined,
- field density spreads,
- configurations approach more homogeneous states.

13.8 Reorganization Dynamics

Following fragmentation or partial dissolution, the system may exhibit reorganization into new structured states.

This behavior depends on:

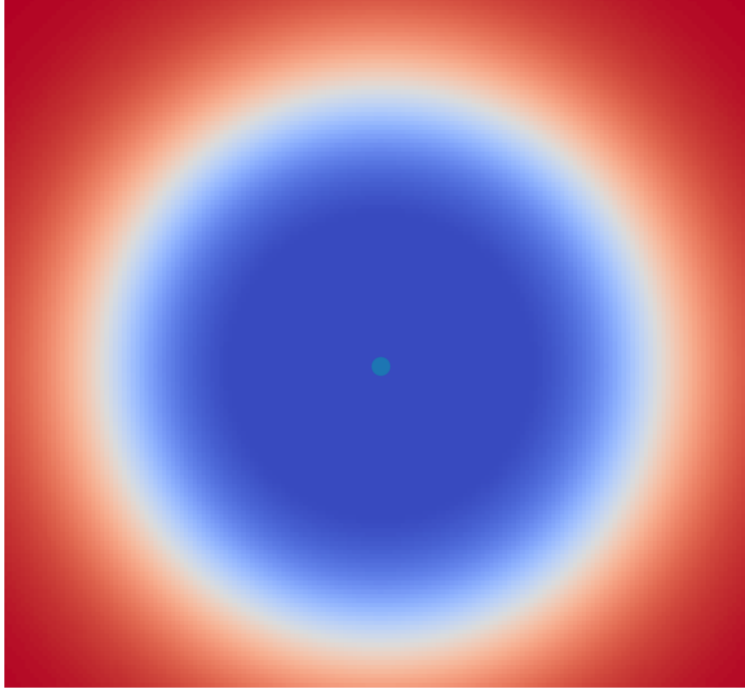


Figure 15: Dissolution of a structured configuration under parameter changes.

- parameter values,
- residual field configuration,
- proximity to the stability window.

We emphasize that such reorganization should be interpreted as a dynamical outcome within the model, and not as evidence of strict cyclic or reversible behavior.

13.9 Transitions Between Regimes

We observe transitions between:

- stable $\rightarrow$ unstable,
- localized $\rightarrow$ extended,
- single-component $\rightarrow$ fragmented.

These transitions depend on:

- time-stepping parameters,
- nonlinear interaction strength,
- coupling parameters.

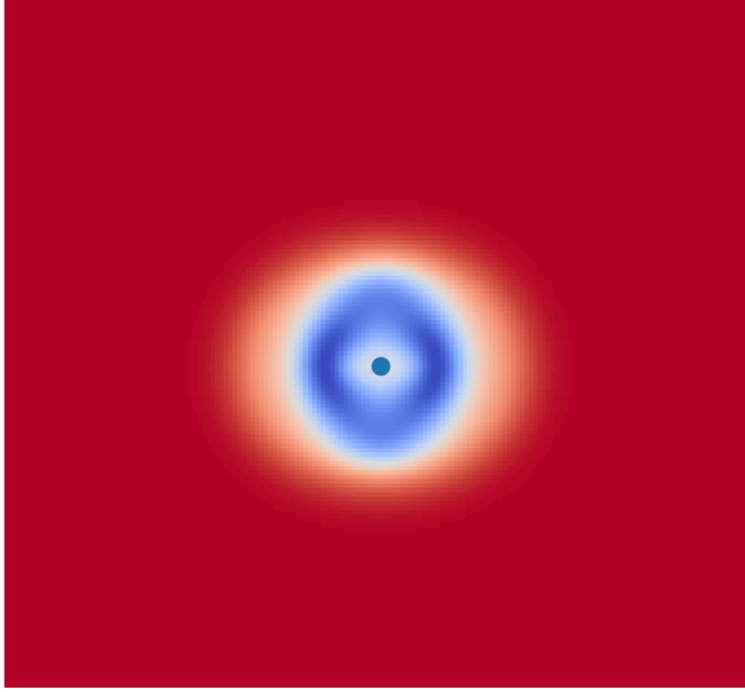


Figure 16: Example of reorganization toward a structured configuration after partial disruption.

13.10 Interpretation

Structure formation, transformation, and dissolution are intrinsic to the nonlinear dynamics of the model.

However, it is essential to distinguish between:

- structural evolution occurring within the numerical stability window,
- numerical artifacts arising from discretization limits.

Only the former should be considered physically interpretable within the scope of this framework.

14 Evolution from Research 1

This work extends and consolidates the framework introduced in Research 1.

14.1 Continuity of the Core Framework

The following foundational principles are preserved:

- a discrete energetic field formulation;
- the emergence of structured topologies;
- an oscillatory substrate interpretation.

These elements constitute the conceptual backbone of both studies.

14.2 Scope and Limitations of Research 1

Research 1 established that non-trivial structured configurations can emerge from a unified computational substrate.

However, several aspects remained only partially explored:

- multi-scale interactions;
- global coherence across extended domains;
- effective large-scale organization;
- systematic reproducibility controls.

As a result, while the emergence of structures was demonstrated, their organization into broader dynamical regimes remained only partially characterized.

14.3 Advances Introduced in the Present Work

The present work addresses these limitations through several extensions:

- the introduction of a two-scale local/global structure;
- the identification of a mantle-like global contribution;
- the analysis of large-scale expansion-like dynamics;
- additional quantitative diagnostics and reproducibility-oriented controls.

These additions extend the structural and interpretive scope of the framework.

14.4 Conceptual Progression

The transition from Research 1 to the present work can be summarized as:

structures $\rightarrow$ regimes

Research 1 established the emergence of structured configurations. The present work investigates how these configurations organize into broader classes of dynamical behavior within the same framework.

14.5 Status of Research 1

Research 1 is not superseded by the present work; it remains foundational.

The current study should be understood as a continuation that explores additional levels of organization, larger-scale structure, and improved numerical characterization within the same underlying model.

15 Unified Interpretation

We summarize the emergent organization of the model through a unified chain linking the different observed levels of description.

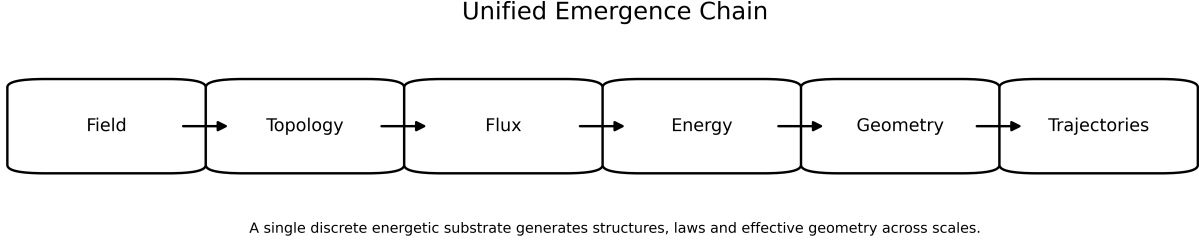


Figure 17: Unified emergence chain of the model: field dynamics generate topological structures, which organize transport, energy distribution, large-scale structure, and effective trajectories.

Field $\rightarrow$ Topology $\rightarrow$ Transport $\rightarrow$ Energy Distribution $\rightarrow$ Large-Scale Structure $\rightarrow$ Trajectories

15.1 Field

The fundamental object of the framework is a scalar field ψ defined on a discrete substrate.
All observed structures and dynamical regimes arise from its evolution.

15.2 Topology

Local variations of the field give rise to coherent topological structures.
These structures represent the first level of organization emerging from the underlying dynamics.

15.3 Transport

Spatial gradients of the field induce effective transport patterns, which govern redistribution of amplitude and mediate interactions between regions.

15.4 Energy Distribution

Energy-like quantities emerge from field amplitude and gradients.
They provide a quantitative bridge between local structure, interaction strength, and dynamical evolution.

15.5 Large-Scale Structure

At sufficiently organized scales, energy distribution affects the broader spatial organization of the field.
This leads to screened regimes, multi-scale organization, and geometry-like effects at larger scales.

15.6 Trajectories

Localized persistent patterns follow trajectories determined by the surrounding field configuration.
In this sense, trajectories are emergent consequences of the spatial organization of the field.

15.7 Interpretive Significance

This chain provides a unified description linking microscopic field dynamics to macroscopic structured behavior.

It suggests that apparently distinct physical-like sectors may correspond to different organizational regimes of a common underlying substrate.

In this perspective, physical-like laws appear as effective descriptions of different organizational regimes of the field.

15.8 Minimal Unified Equation

A compact representation of the multi-scale structure is:

$$(\nabla^2 - m_\ell^2)\psi + \eta(\nabla^2 - m_g^2)\psi = \rho \quad (31)$$

This expression encodes the interplay between local screened dynamics and global coupling contributions.

It should be understood as a compact structural summary, not as a fundamental field equation derived from first principles.

Remark on measurement-dependent correlations

Additional exploratory diagnostics, including Bell-like correlation constructions, were investigated in separate numerical studies.

These analyses highlight the sensitivity of measured correlations to observable definitions and reconstruction procedures.

However, these results remain exploratory and are not part of the core validation of the present framework.

15.9 Interpretation and Limitations

The present framework should be understood as an exploratory and emergent modeling approach rather than a fully derived physical theory.

The correspondences with known physical equations (Schrödinger-like, Dirac-like, transport-like, geometry-like structures) are to be interpreted as effective and phenomenological.

In particular:

- no formal derivation of these equations is claimed;
- observed similarities arise from structural and dynamical analogies;
- the model operates at a level of abstraction where multiple physical interpretations remain possible.

These results should therefore be viewed as indications of structural compatibility rather than strict equivalence.

The system exhibits sensitivity to initial conditions, as expected in nonlinear dynamical systems.

At the same time, several structural tendencies and regime classes remain reproducible across controlled simulations.

16 Predictions and Falsifiability

A unified framework is scientifically meaningful only if it produces distinguishable predictions and can, in principle, be falsified.

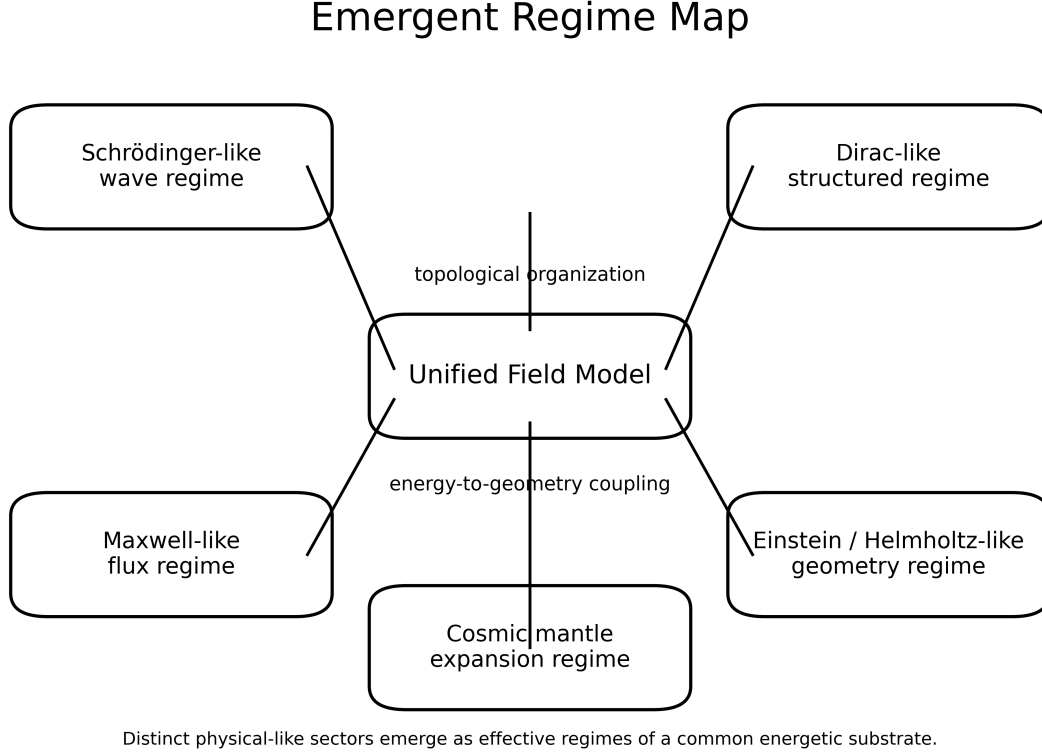


Figure 18: Regime map of the unified framework: wave-like, structured, transport-like, geometry-like, and expansion-like behaviors appear as effective sectors of the same underlying field dynamics.

16.1 General Principle

The present model predicts that multiple structured regimes can arise from a single energetic substrate through scale-dependent organization.

These predictions are formulated in terms of observable properties of the simulated field and can, in principle, be tested through systematic numerical exploration across parameter regimes.

16.2 Prediction 1: Scale-Dependent Interaction Regimes

The model predicts the coexistence of at least two interaction regimes:

- a local screened regime, characterized by short-range behavior;
- a global coupled regime, extending the spatial range of organization.

This implies that effective interaction range may depend on the configuration and scale of the field.

16.3 Prediction 2: Emergent Screening Mass

The model predicts the existence of an effective screening parameter

$$m_{\text{eff}}^2 = m_\ell^2 + \eta m_g^2 \quad (32)$$

which depends on the balance between local and global scales.

This implies that screened and less-screened regimes may correspond to different structural states of the same substrate.

16.4 Prediction 3: Nonlinear Expansion-like Spreading

The large-scale numerical scans indicate that the model can produce enhanced spreading relative to diffusion.

A testable prediction is therefore that, for suitable parameter ranges, nonlinear regimes generate larger effective scale-factor growth than purely diffusive controls.

At present, this prediction concerns structured nonlinear spreading, and does not constitute a validated cosmological mechanism.

16.5 Prediction 4: Structural Transformation Regimes

The model predicts that structures are not permanently fixed. Depending on the parameter regime, they may:

- remain stable;
- fragment;
- dissolve;
- reorganize into new structured states.

This indicates that organization in the model is intrinsically dynamical.

16.6 Prediction 5: Regime Transitions Across Sectors

Because the same substrate generates wave-like, localized, transport-like, and large-scale organized regimes, the model predicts the existence of transition thresholds between these sectors.

These thresholds correspond to changes in structural organization as system parameters vary.

16.7 Falsifiability Criteria

The framework would be weakened or falsified if systematic numerical exploration showed that:

- distinct regimes cannot be recovered from a common set of governing dynamics;
- the observed structural classes are numerically unstable or non-reproducible;
- the multi-scale structure fails to produce coherent local/global organization;
- apparent regime differences collapse under improved diagnostics or controls.

16.8 Current Epistemic Status

At the present stage, the framework should be regarded as:

- numerically supported;
- structurally unified;
- qualitatively predictive;
- not yet experimentally validated.

16.9 Extreme Regimes

The framework suggests that extreme concentrations of energy may generate highly localized and strongly deformed regimes.

Whether such regimes admit robust analogies with trapping or horizon-like behavior remains an open question requiring dedicated investigation.

16.10 Outlook

The most direct next step is to confront the model with more explicit numerical and physical targets, including:

- localized bound structures;
- magnetic interaction patterns;
- orbital or quasi-orbital dynamics;
- large-scale spreading profiles.

16.11 Beyond Large-Scale Regimes

While large-scale organization provides one important interpretation of the model, the same framework also produces a broader spectrum of lower- and intermediate-scale structures.

These include:

- localized multi-component structures;
- proto-atomic and family-like configurations;
- composite systems with structured transport exchange;
- dipolar and interaction-like regimes;
- orbital and quasi-orbital trajectories.

These regimes provide a direct basis for further falsifiable numerical investigation.

16.12 Additional prediction: high-energy nonlinear dispersion

Numerical observations suggest that, at high wave numbers, the effective dispersion relation deviates from linear behavior.

In particular, deviations consistent with a cubic correction:

$$E(k) \sim k + \gamma k^3 \quad (33)$$

may emerge in strongly nonlinear regimes.

This effect, if confirmed and quantified, would provide a potential falsifiable signature of the underlying discrete nonlinear dynamics.

At present, this observation remains preliminary and requires further numerical validation.

This deviation is not imposed in the model, but arises spontaneously from the underlying dynamics.

17 Conclusion

We have introduced a computational framework in which multiple classes of structured dynamical behavior emerge from a single discrete scalar field.

The results show that wave-like propagation, localized structures, transport-mediated redistribution, and large-scale organization can arise within a unified dynamical system governed by multi-scale coupling and nonlinear interaction.

A key contribution of this work is the identification of a two-scale structure combining local screened dynamics and a global coupling component. This structure provides a consistent mechanism through which both short-range and extended behaviors can coexist and interact.

Across a range of parameter regimes, the model exhibits robust formation of structured configurations and transitions between distinct dynamical regimes. These behaviors are reproducible under controlled numerical conditions and can be partially characterized using quantitative structural descriptors.

The correspondences observed with known classes of physical behavior should be interpreted as structural and phenomenological. No formal derivation of established physical theories is claimed.

At the present stage, the framework remains exploratory and numerically grounded. Further work is required to:

- establish quantitative connections with known physical systems;
- analyze parameter universality and scaling behavior;
- improve numerical schemes and long-time stability;
- develop reproducible benchmarks and independent validations.

The results suggest that complex structured behaviors can emerge from relatively simple underlying dynamics through multi-scale organization.

This work therefore provides a candidate framework for investigating how diverse classes of dynamical behavior may arise within a unified computational substrate.

This framework is intended as a starting point for further analytical and experimental investigation rather than a final physical theory.

18 Reproducibility

All simulations and results presented in this work are reproducible using the accompanying open-source codebase, available at:

<https://github.com/NicolasGilbertAlbertRoux/OMM-SBT.git>

The repository provides:

- a modular implementation of the unified field model;
- scripts for reproducing the main regimes discussed in the paper;
- pipelines for generating the principal figures;
- selected final states and datasets for validation and inspection;
- a structured organization separating core dynamics, diagnostics, and domain-specific regimes.

A unified entry point allows reproduction of the main results:

```
python main.py --target all
```

Alternatively, individual components can be reproduced separately:

```
python main.py --target chain
python main.py --target regime-map
python main.py --target lifecycle
python main.py --target mantle
python main.py --target cosmo
```

All numerical outputs are reproducible given identical parameters, initial conditions, and random seeds (when applicable), and intermediate data are saved to enable detailed inspection.

The codebase is designed to be:

- reproducible: the principal figures can be regenerated from the provided scripts;
- transparent: intermediate states are accessible for verification;
- extensible: additional regimes and parameter explorations can be added.

This structure allows independent verification, reproduction of results, and further exploration of the model by the research community.

Reproduction of the main figures and regimes can typically be achieved on standard personal computing hardware (CPU-based execution).

18.1 Reviewer-Oriented Numerical Controls

In addition to the main reproduction pipeline, the public repository includes targeted methodological controls addressing:

- explicit nonlinear potential choice;
- regime distinction metrics;
- cosmology-versus-diffusion comparison;
- stability tests with and without clipping.

These controls are provided to clarify the numerical scope of the model and to distinguish exploratory safeguards from stability-controlled runs.

18.2 Supplementary Bell-like Validation Scripts

The public repository also includes supplementary validation scripts dedicated to Bell-like and event-reconstruction diagnostics.

These scripts are not part of the main evidential pipeline of the paper, but are provided as methodological stress tests for observer-side event attribution, anisotropic preparation, and bounded versus unbounded correlation aggregation.

In particular, the repository includes:

- an anisotropic-preparation baseline scan;
- an extended normalized event-tracking scan;
- a targeted event-tracking validation used to assess robustness of Bell-like excesses under bounded normalization.

Additional Lorentz–Pauli-oriented diagnostic scripts may also be included as supplementary calibration tools.

These components document how apparent Bell-like excesses can depend on reconstruction choices and support future refinement of calibration procedures across the broader framework.

These diagnostics are exploratory and should not be interpreted as evidence of fundamental violations of established physical principles.

A Correspondence Between Model Quantities and Physical Concepts

The table below summarizes the main quantities introduced in the model and the qualitative concepts with which they are associated in the present work.

Model Quantity	Qualitative Interpretation
ψ	scalar field (model variable)
$\nabla\psi$	gradient-based transport proxy
$ \psi ^2$	energy-density proxy
local screening m_ℓ	short-range screening scale
global coupling scale m_g	large-scale organization scale
η	coupling between local and global contributions
topological structures	persistent organized configurations

These correspondences are descriptive and should not be interpreted as exact identifications with established physical observables.

B Relation to Known Classes of Equations

The framework displays regimes that are structurally reminiscent of several well-known classes of equations used in mathematical physics.

Standard Equation Class	Qualitative Structural Relation
Schrödinger-type	wave-like regime of ψ
Dirac-type	localized multi-component structured regime
Maxwell-type transport	flux-organized regime
Helmholtz / Yukawa	screened local interaction regime
geometry-like dynamics	organization influenced by energy distribution

These relations are phenomenological and structural. They are included to situate the observed regimes, not to claim formal equivalence.

C Magnetic, Atomic, Molecular and Orbital Illustrations

This appendix provides illustrative numerical examples of emergent structures obtained within the unified framework across different organized regimes.

These examples are included as qualitative demonstrations of the range of structures supported by the model.

C.1 Localized and Proto-Atomic Structures

Localized multi-lobed configurations emerge naturally and exhibit persistent internal organization.

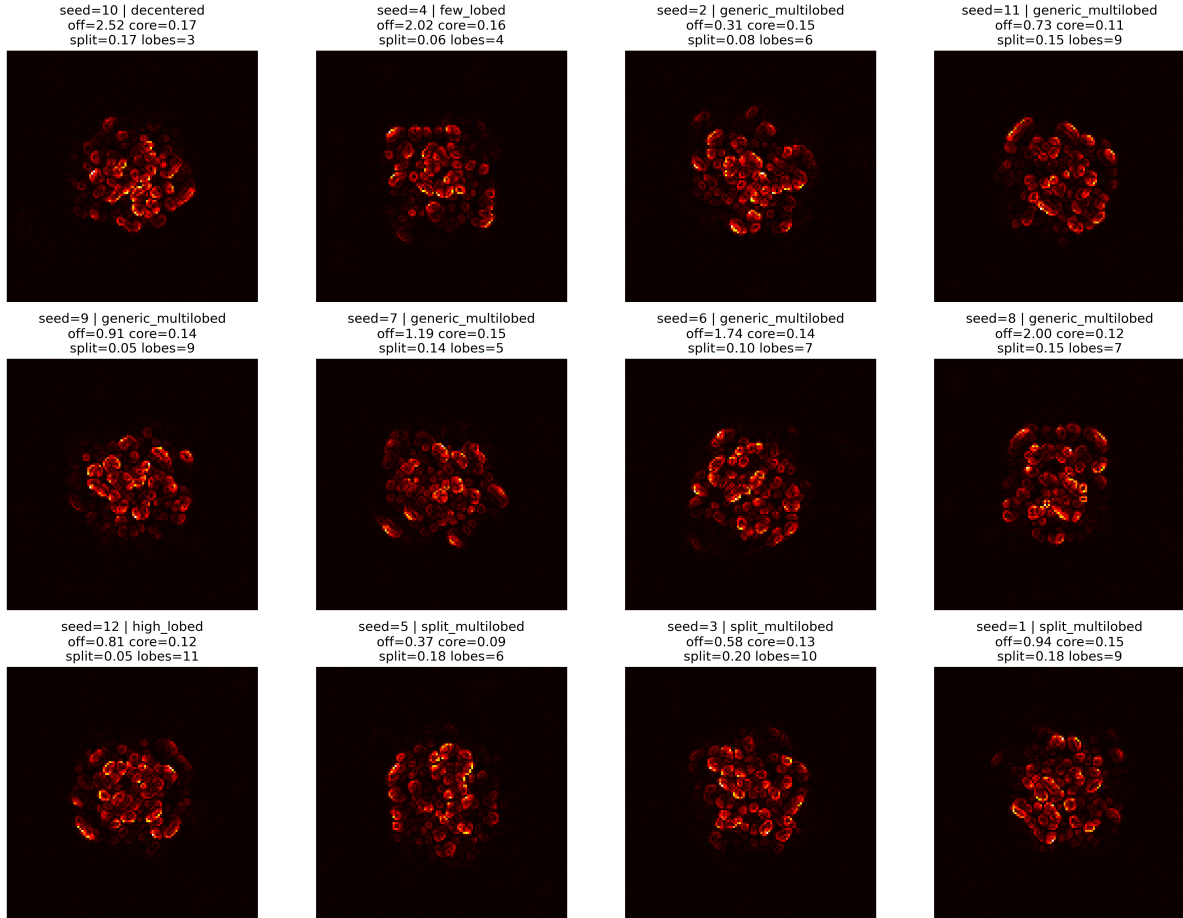


Figure 19: Examples of localized and proto-atomic structured configurations.

C.2 Flux and Node Organization

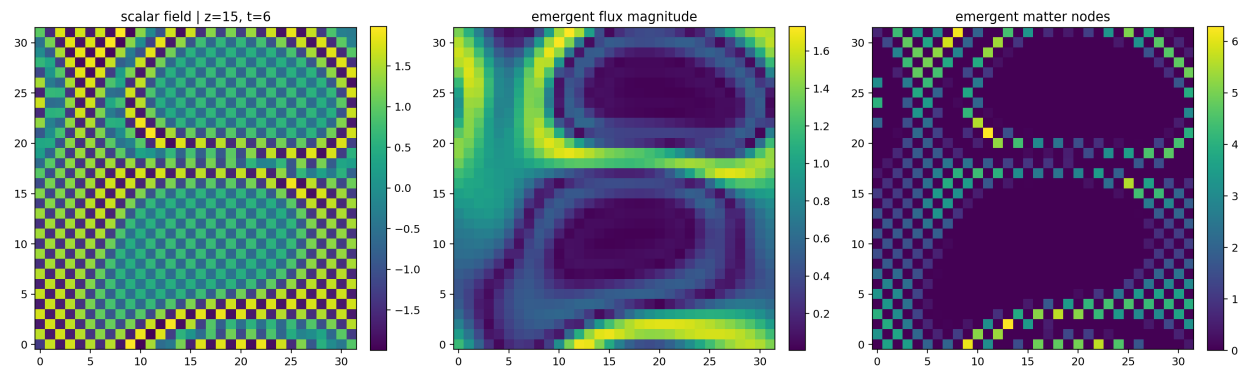


Figure 20: Field, flux magnitude, and node distribution illustrating structured organization.

C.3 Dipolar Interaction Regimes

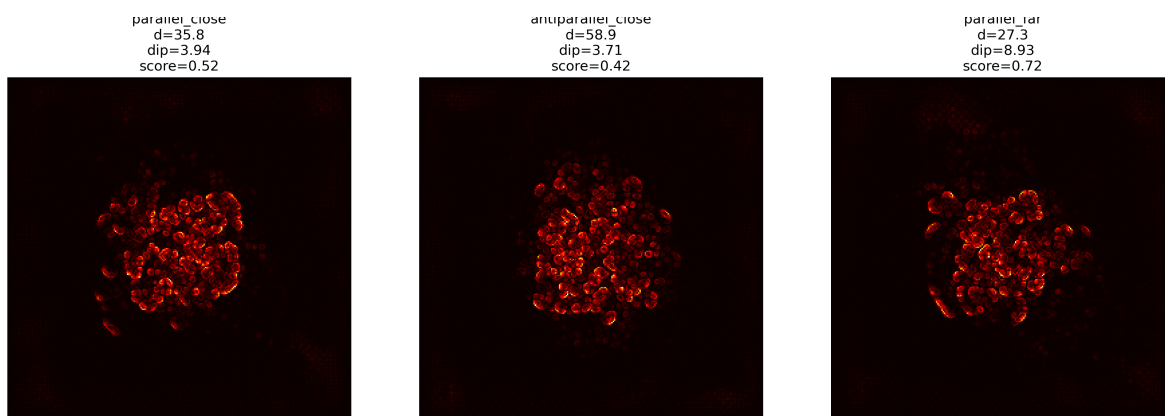


Figure 21: Examples of dipolar interaction regimes with orientation-dependent behavior.

C.4 Effective Dipole Structure

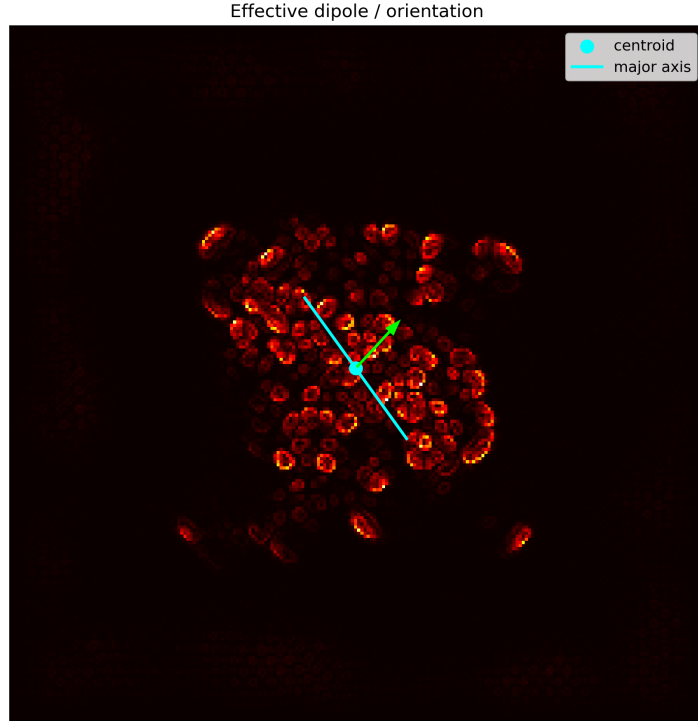


Figure 22: Example of an effective dipole extracted from an emergent structure.

C.5 Orbital and Collective Dynamics

Additional simulations indicate that structured trajectories and collective motion can emerge under suitable conditions.

These examples remain qualitative and are provided as illustrations of the broader exploratory scope of the framework.

D Descriptor-Based Regime Comparison

To address the question of regime distinguishability beyond visual inspection, we performed a descriptor-based comparison across several representative regimes.

Each regime is characterized using structural metrics including: kurtosis, participation ratio, active fraction, radial concentration, angular anisotropy, and ring contrast.

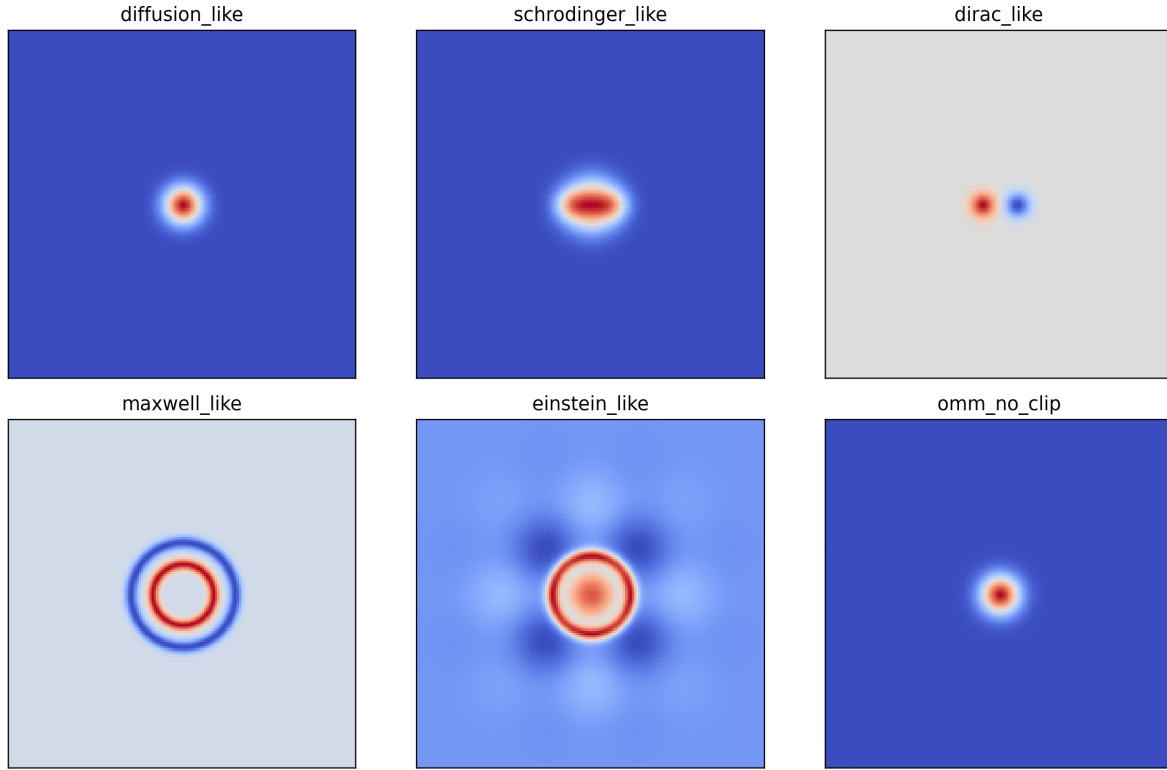


Figure 23: Representative field configurations used for regime comparison. Each panel corresponds to a distinct structural regime: diffusion-like, Schrödinger-like, Dirac-like, Maxwell-like, Einstein-like, and a no-clipping OMM run.

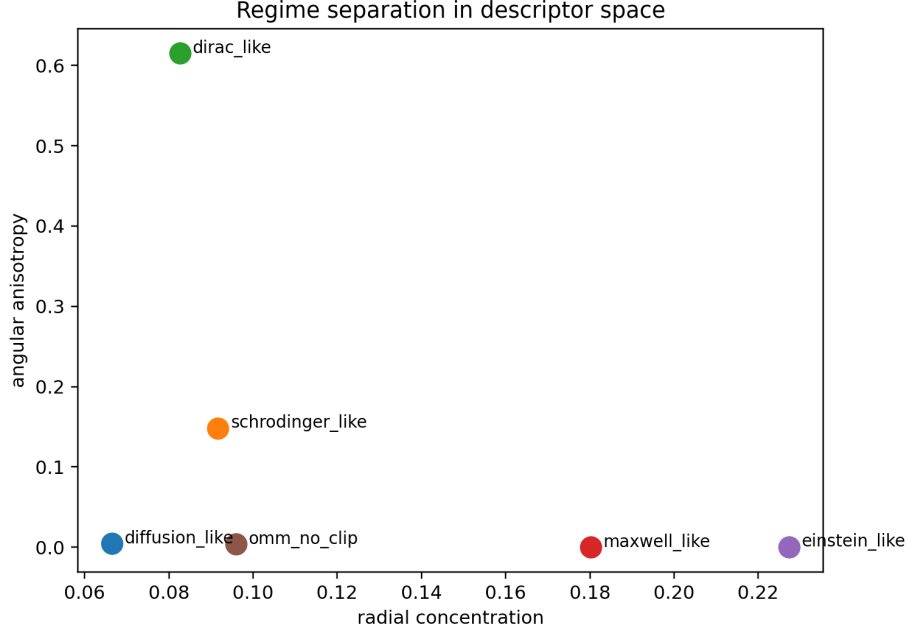


Figure 24: Separation of regimes in descriptor space. Distinct regimes occupy different regions, indicating structural differentiation beyond simple amplitude scaling.

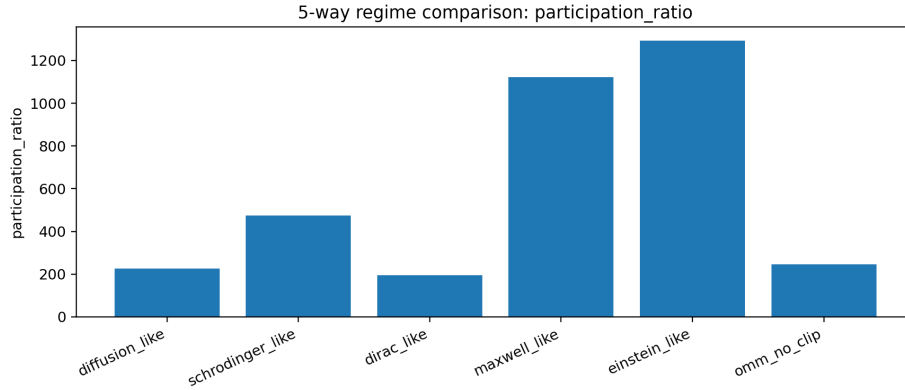


Figure 25: Comparison of participation ratio across representative regimes, highlighting differences in spatial localization and structural distribution.

The results indicate that the observed differences are not reducible to amplitude rescaling alone, but correspond to genuine differences in structural organization.

At the same time, these comparisons remain numerical and descriptor-based: they do not establish formal phase boundaries or a complete classification theorem.

E Emergent Structural Atlas and Proto-Periodic Classification

The unified framework produces a broad range of structured configurations that can be organized into families based on topology, symmetry, stability, and interaction behavior.

E.1 Classification Criteria

Structures are grouped according to:

- topology;
- symmetry properties;
- persistence over time;
- interaction behavior.

E.2 Families of Structures

The simulations reveal several recurring families, including:

- localized nodes;
- oscillatory shells;
- dipolar configurations;
- extended flux structures.

E.3 Proto-Periodic Classification

These families exhibit recurring organizational motifs suggesting a proto-classification of structures.

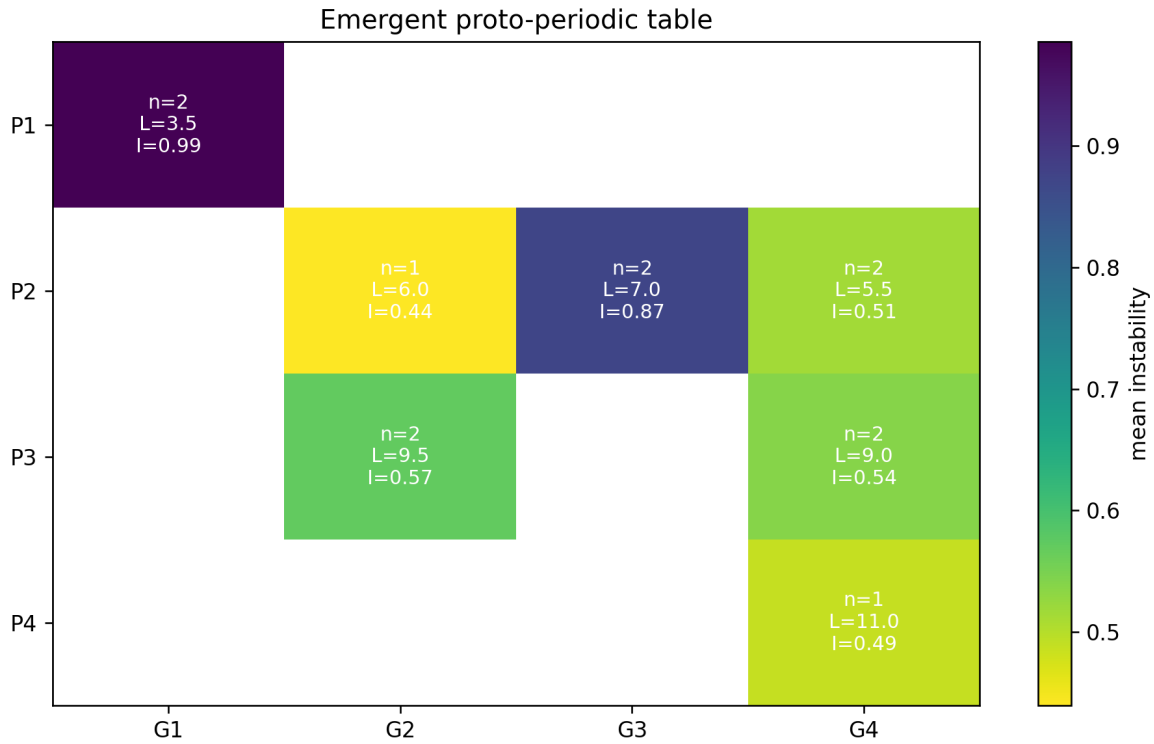


Figure 26: Illustrative proto-classification of emergent structures based on topology and persistence.

E.4 Interpretation

This organizational atlas is intended as a descriptive classification tool.

It suggests that structural diversity may arise from variations within a common generative mechanism, but it should not be interpreted as a literal analogue of the chemical periodic table.

F Illustrative Simulations from the Unified Model

The unified framework enables the simulation of configurations that qualitatively resemble several classes of organized physical-like systems.

F.1 Atom-like Visualization

Some simulations produce localized structures with internal oscillatory organization, motivating atom-like visual interpretations.

F.2 Molecule-like Flux Organization

Interactions between multiple structures can produce composite configurations with structured flux exchange.

F.3 Multi-Interaction Regimes

Flux-mediated interactions between extended structures can produce orientation-dependent attraction and repulsion patterns.

F.4 Orbital and Quasi-Orbital Trajectories

Under suitable conditions, stable or quasi-stable trajectories emerge from the organized field background.

These examples are illustrative and exploratory. They document the diversity of behaviors supported by the framework, without claiming direct derivation of the corresponding physical systems.

G References

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